



Hands on Homework #2 (Particle Equilibriums-Springs)

EGN 3310 - 0M01 Engineering Analysis: STATICS

MWF, 11:30AM-12:20AM

Submitted by: Group 38

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Graphical Worked Out on Board

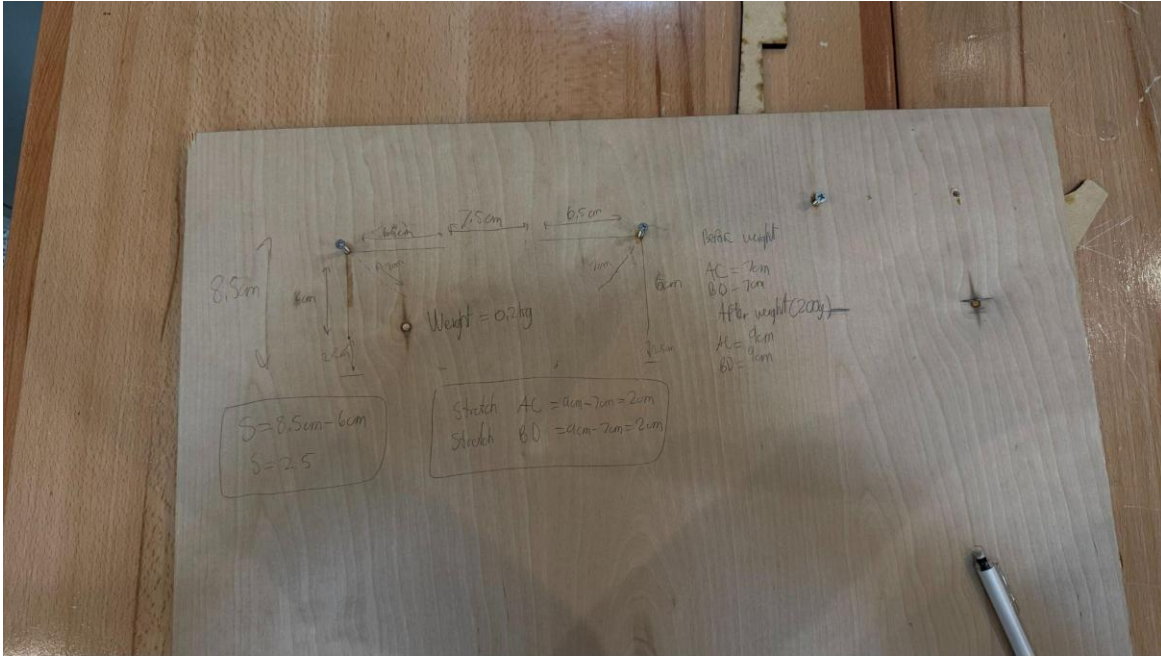


Fig. 2.1: picture of model before springs (rubberbands) are attached

Hands-On Method (Physical Model)

For the hands on model, we had to draw everything out and follow the model we used for our NID.

We decided to adjust the values in order to make it feasible for use on the board. Also every measurement will be converted to cm's, including the spring constant.

Please refer to Fig. 2.2 below for the new distances.

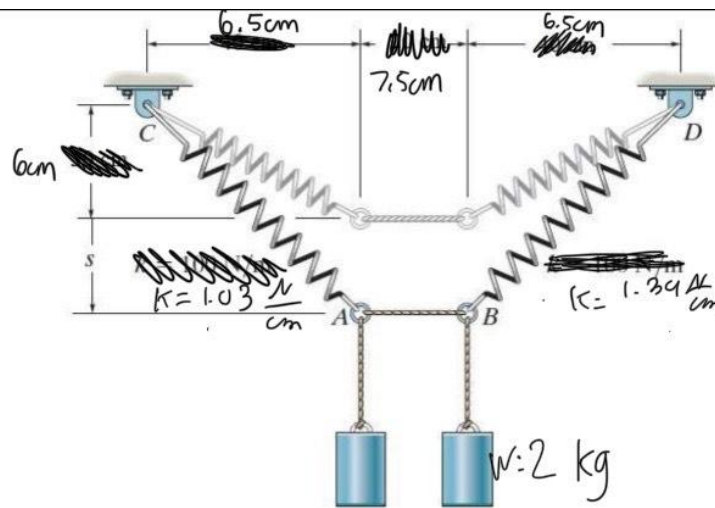


Fig. 2.2: diagram of our model with our actually measurements

In the image above, the value of the spring constants (which in our case is the rubber bands) was found by hanging a weight on the unstretched rubber band and seeing the distance that it was stretched by. We applied a small weight of 50g to the rubber band as Dr. Zaurin did his video to make the starting point. Following that we added mass to that to calculate the distance stretched.

Using $F_{spring} = k * x$, we are able to find the spring constant.

Of course multiple weights were used for each rubber band to reduce error as much as possible. We found two rubber bands that had the same or very similar stretch distance, as it is required.

Here are pictures of the stretched distances

Spring 1



Fig. 2.3-2.6: Physical measurements used for calculating spring constant k for spring (rubberband) AC

Spring 2

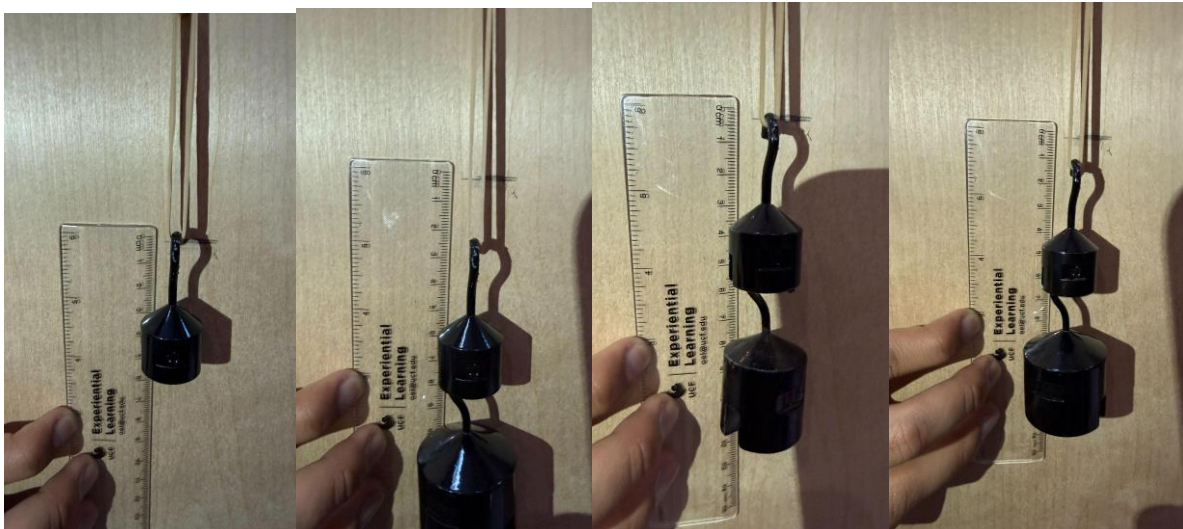


Fig. 2.7-2.10: Physical measurements used for calculating spring constant k for spring (rubberband) BD

Below is a graph and table with the stretch and the force, and the average we found for the spring constant. These graphs give us the spring constants for each of our two rubber bands in N/cm.

Force (N) vs. Stretch (cm) ("Spring" 1)

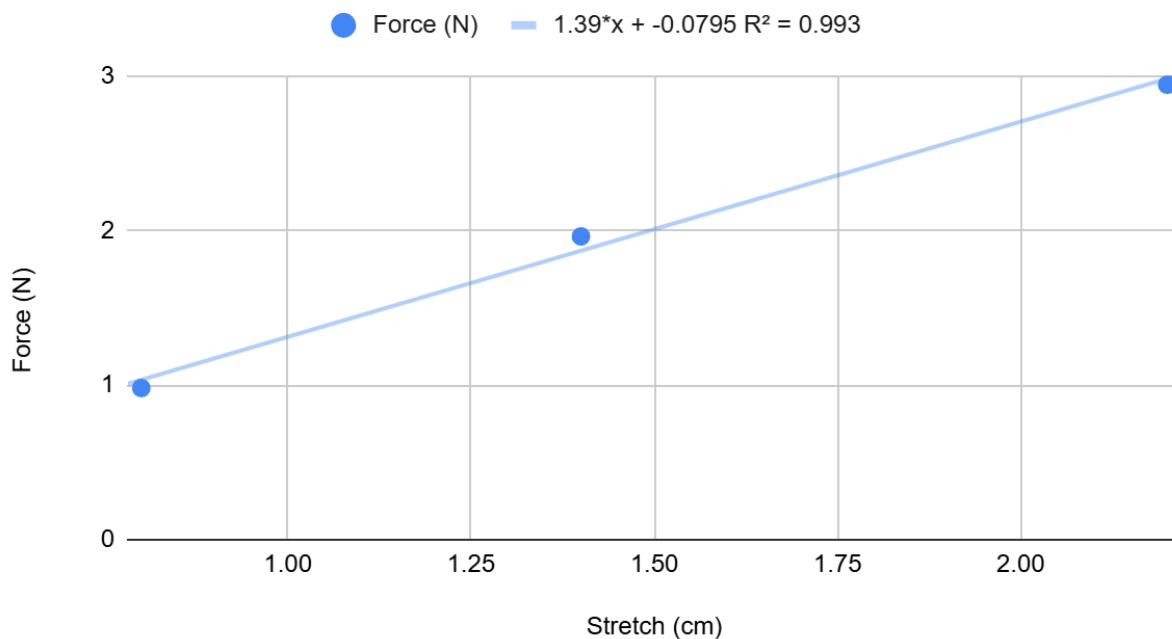


Fig 2.11: Graph showing the spring constant for “Spring” 1 (BD)

This first graph gives us the spring constant for the rubber band through the equation of the average of all three weights. $1.39x$, where 1.39 is k in N/cm.

The original problem had a spring constant of 100N/m , which means we have some error, as our k value converted to meters gives us 139 N/m

Here is the percent error:

$$\left(\frac{|139 - 100|}{100} \right) \times 100\% = 39\%$$

Force (N) vs. Stretch (cm) ("Spring" 2)

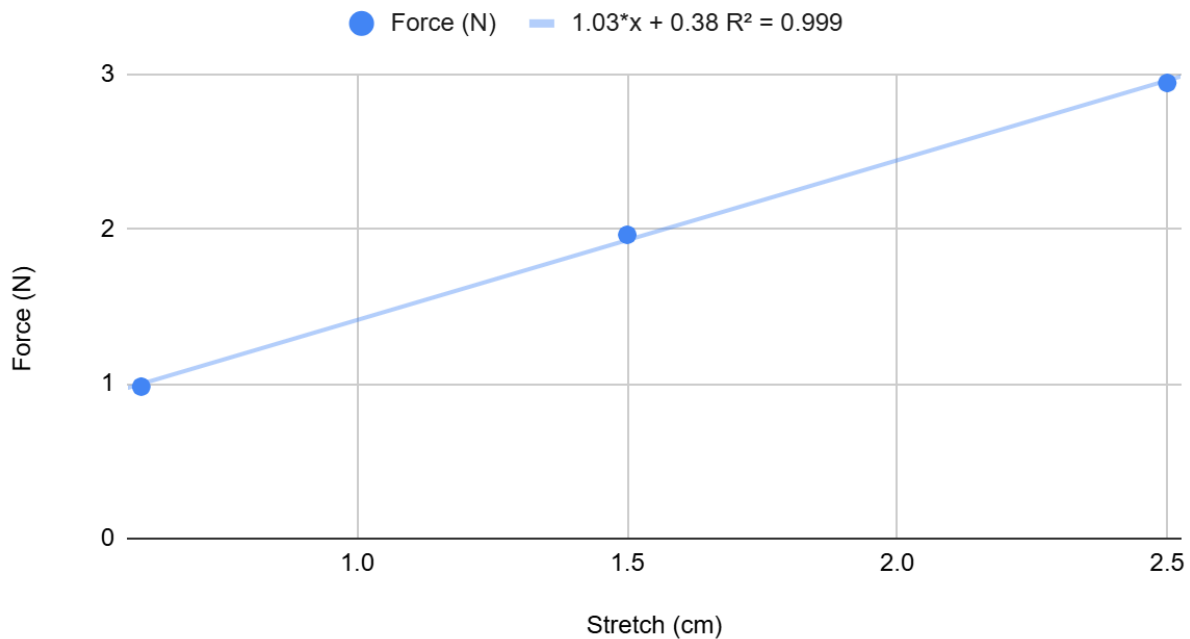


Fig 2.12: Graph showing constant for “Spring” 2 (AC)

This second graph gives us the spring constant for the rubber band through the equation of the average of all three weights. $1.03x$, where 1.03 is k in N/cm .

Once again some error can be seen, which is expected. However this one is less than the previous one, yielding (when converted) a spring constant of 103N/m .

Here is the percent error:

$$\left(\frac{|103 - 100|}{100}\right) \times 100\% = 3\%$$

Finally, here is the exact same model as in the original document both without the weights and with the weights.



Fig. 2.13: picture of our model with the 2 200g masses

Fig. 2.14: picture of our model without any masses

After doing the experiment, we were able to find the value for s , as well as the stretch for both AC and BD. The masses we used are 200g or .2kg. This means the weight is $9.81 \times 0.2 = 1.96N$

Please refer to the first image at the top for the distance calculations, but here is a summary.

Using the unstretched lengths of the rubber bands and then the stretched ones with the weights on it, we could find the total stretch of the bands.

$$\text{Stretch } AC = 9cm - 7cm = 2cm$$

$$\text{Stretch } BD = 9cm - 7cm = 2cm$$

Then for s , we take the vertical values from the supports at the top to the center of each ring when it is unstretched, and subtract that from when they are stretched.

$$s = 8.5cm - 6cm = 2.5cm$$

That concludes the hands-on model. We were already given the weights as we could choose them, but we were able to find the spring constant for each rubber band, as well as their stretch and s .

Now we have data to compare to for the hand calculation method.

Hand Calculation (Theoretical Method)

Here is once again the original problem with the adjusted values and all the calculations. These will be the values used to calculate what the questions asks (weight, and stretch), and we can compare that to our hands on model.

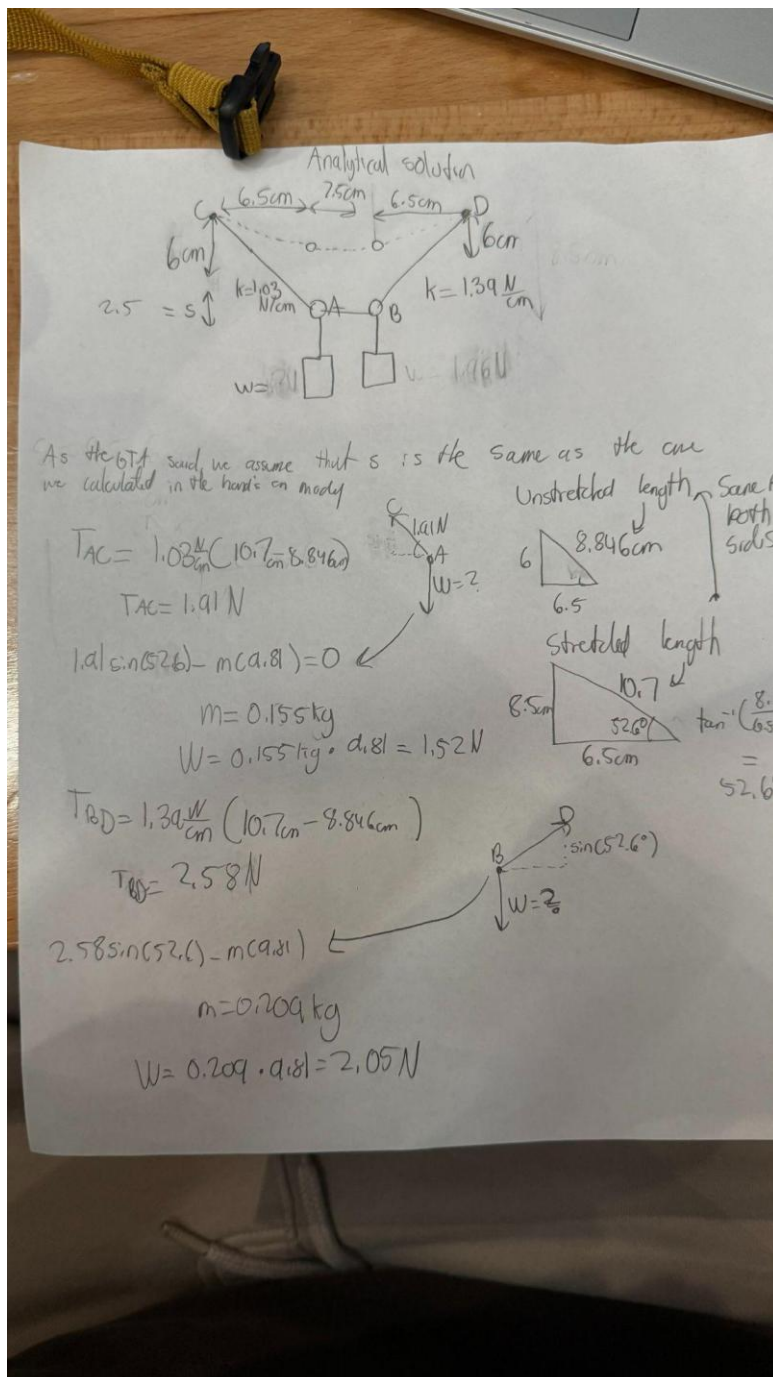


Fig 2.14: Picture of our Analytical solution on paper

For the hand calculation portion, we used the same measurements as the ones we measured in the hands-on portion, as that is required to find the error we had. This means the spring constants, the distances and s are the same (The GTA told us to keep the same). We now need to find the stretch distance for **AC** and **BD** as well as the weight of the block.

Stretch AC and BD

For the stretches, we had to use pythagorean theorem to find the length of both “springs” in both their unstretched and stretched positions.

Since the distance in the x-direction is 6.5cm for both their unstretched and stretched portion of both springs, that solves the horizontal portion of the triangle. We also know from the hands-on model that the uncompressed vertical stretch is 6cm, and the total stretch is 8.5cm. Now we can make a triangle and solve using pythagorean theorem for the unstretched and stretched values.

$$\text{Unstretched length} = \sqrt{(6)^2 + (6.5)^2}$$

This gives us a value of 8.846 cm for the unstretched length

$$\text{Stretched length} = \sqrt{(6.5)^2 + (8.5)^2}$$

This gives us a value of 10.7cm for the stretched length

For the angle, we had to use $\tan^{-1} = \frac{8.5}{6.5} = 52.6^\circ$. We used the values of the stretched lengths.

These values for the unstretched and stretched lengths are the same for both **AC** and **BD**, since the values we used above were calculated for both. This is a probable source of error for the results we got in the following sections.

This means our stretched distance is $10.7\text{cm} - 8.846\text{cm} = 1.854\text{ cm}$ for both **AC** and **BD**.

	Unstretched	Stretched	Difference
Hands-on	7cm	9cm	2cm
Hand-Calculated	8.846cm	10.7cm	1.854cm

Fig. 2.15: Table of measurements of the actual stretched and unstretched lengths measured in centimeters, as well as the calculated lengths

Now for the percent error:

We calculated that unstretched length in the hands-on was 7cm and the stretched was 9 cm. $9\text{cm} - 7\text{cm} = 2\text{cm}$, which is different from the one we calculated.

$$\text{Percent error} = \left| \frac{1.854 - 2\text{cm}}{2\text{cm}} \right| \times 100\% = 7.3 \%$$

Tension AC and weight of mass on AC:

For finding the tension of AC, we used the spring constant we calculated in the hands-on model using the different weights.

The values for the angles came from the section above, and we used arctan to find it.

After calculating for the stretched and unstretched length, we can now calculate the mass and tension using the formula of Hooke's law ($F \propto kx$):

Given the numbers from stretched and unstretched length:

$$T_{Ac} = (1.03 \frac{N}{cm})(10.7\text{cm} - 8.846\text{cm})$$
$$T_{Ac} = 1.91N$$

Now calculating for mass and weight:

$$1.91\sin(52.6) - m(9.81) = 0$$
$$m = 0.155\text{kg}$$

$$w = 0.155\text{kg} * 9.81 = 1.52\text{N}$$

	Mass	Weight
Hands-On	.2kg	1.96N
Hand-Calculated	.155kg	1.52N

Fig. 2.16: table of masses and weights for “spring”/tension AC

Now given our value, we calculate for our percent error:

$$\text{Percent Error} = \left(\frac{|1.96 - 1.52|}{1.52} \right) \times 100\% = 28.9\%$$

Where,

1.96 was our expected weight while 1.52 was our actual calculated weight.

Tension BD and weight of mass on BD:

For finding the tension of BD, we used the spring constant we calculated in the hands-on model using the different weights.

The values for the angles came from the section above, and we used arctan to find it.

For tension BD, we follow the same process as finding Tension AC.

$$T_{BD} = 1.39 \left(\frac{N}{cm} \right) * (10.7cm - 8.846cm)$$
$$T_{BD} = 2.58N$$

Now, we find the weight of mass on BD.

$$2.58 \sin(52.6^\circ) - m(9.81) = 0$$

$$m = 0.209kg$$

$$W = 0.209 * 9.81 = 2.05N$$

	Mass	Weight
Hands-On	.2kg	1.96N
Hand-Calculated	.209kg	2.05N

Fig. 2.17: table of masses and weights for “spring”/tension BD

Percent Error of weight on BD:

$$\text{Percent Error} = \frac{|2.05 - 1.96|}{1.96} * 100\% = 4.59\%$$

Where,

1.96 was our expected weight while 2.05 was our actual calculated weight.

Explanation of Percent Errors:

As for the percent error in the stretched lengths of the rubber bands, the error most likely stemmed from the fact that measuring the length of both the stretched and unstretched perfectly in the hands on is hard due to obstacles in the way as well as human error when looking at the measurements. Even just .1 cm difference causes a great error.

For calculating the percent error for the weights on BD and AC, we made a realization that even though “spring” BD had significantly more percent error in the spring constant than “spring” AC, the percent error for weight on BD was significantly less than on AC.

One large factor that could’ve played a role was the use of large ring hooks (which was all we had) in our model instead of using smaller ones. This could’ve changed the actual measurements of the distance of the stretch as well as x and y components of each rubberband. As a result, this would’ve changed the angles formed, and thus changed how we calculated the mass and weight on BD and BC because of the use of $\sin()$ in our calculations.

Conclusion/discussion:

Vincent Tremblay (vi474918):

Overall, even though we are utilizing rubber bands for the experiment, our data was not too skewed.

Of course, most of the data depended on the spring constant we measured at the start, which could not be accurate due to the nature of rubber bands VS springs. Even so, we were able to effectively measure the distance stretched for both “springs” as well as the weight for each side through the use of static principles, leading to a successful learning experience for us.

Andrew Tran (an365172):

Considering the amount of potential error, mainly from the lack of proper equipment and having to use substitutes, our results analytically were relatively close to the actual model (especially with the weight on “string” BD having low percent error). Because of this, I believe our model is good evidence of basic physics concepts such as Hooke’s Law $F = kx$. If we had used smaller ring hooks, I believe our analytical results would’ve been even closer to the real model.

Kevin Sanchez Ramos (ke987766):

All things considered, after all our measurement, literally using items in our pockets, the data comparison from the graphical and analytical approach we were able to show that our measure of accuracy was decent enough to prove the uses of real world application of creating a somewhat accurate experiment of data. Though some percent error was expected, it still showed we were capable of success despite some road blocks.